



Virtual Internship

Cleaning & Analyzing Employee Dataset

Task 1

01

Data Cleaning

5 requests in the raw dataset , fixed one by one in Power Query

Attendance_data.csv | Power BI | Power Query

1.Check for duplicates in the dataset and remove them.

Problem

Total Rows before removing duplicates → 6336

	date	weekno	employee_id	name	status
	Valid100% Error0% Empty0% 13 distinct, 0 unique	Valid100% Error0% Empty0% 3 distinct, 0 unique	Valid100% Error0% Empty0% 78 distinct, 12 unique	Valid100% Error0% Empty0% 158 distinct, 1 unique	Valid100% Error0% Empty0% 6 distinct, 0 unique
1	01-04-2022	W 14	Atq -458@	JESSIE CARRILLO	Work From Office
2	01-04-2022	W 14	Atq -400	chad macias	Work From Office
3	01-04-2022	W 14	Atq -437	dallas weeks	Work From Office
4	01-04-2022	W 14	Atq -369	GRETA HORTON	Work From Office
5	01-04-2022	W 14	Atq -374	KALEY FITZPATRICK	Work From Office
6	01-04-2022	W 14	Atq -470	JOVANI POPE	Work From Office
7	01-04-2022	W 14	Atq -455	caylee meadows	Work From Office
8	01-04-2022	W 14	Atq -385	FINNEGAN GILBERT	Work From Office
9	01-04-2022	W 14	Atq -387	KONNER HENSON	Work From Office
10	01-04-2022	W 14	Atq -443	andrew cummings	Work From Office

Approach

Home > Transform Data > Left click on “date column” > Shift Key + Left click on “Status Column” > Right Click > Remove Duplicates

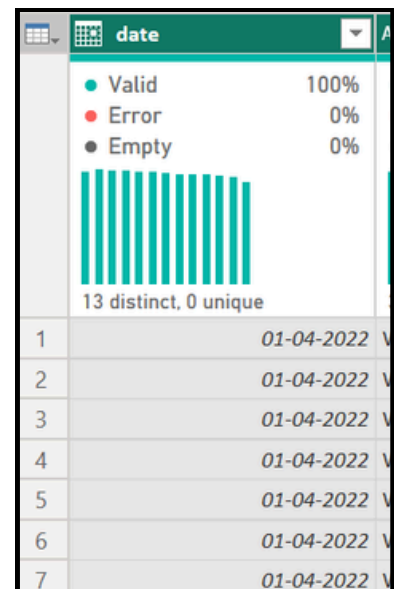
Result

Total Rows After removing duplicates → 6208

2. Standardize the date values to the format YYYY-MM-DD and extract the month name and day type from them.

Problem

Date formatted dd-MM-yyyy

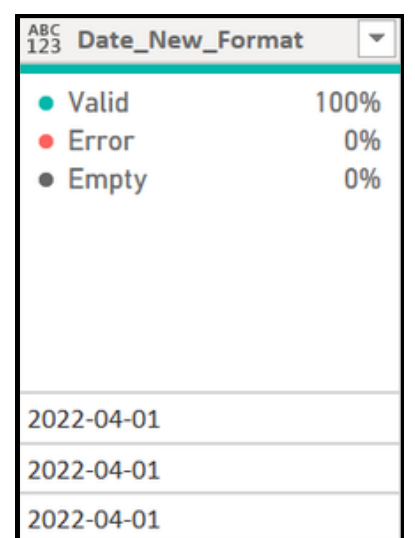


Approach

First Duplicated "date column" > Then navigated to Add column ribbon > clicked custom column > entered Column name "Date-New-Format" and entered the following as custom column formula
`=Date.ToText([date],"yyyy-MM-dd")`

Result

Date formatted into yyyy-MM-dd

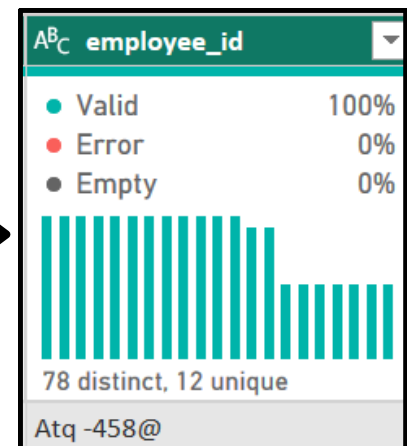


- Remaining Requests Duplicated date column > Then Home Ribbon > Clicked Date & Time Column > Month > Name of the Month
- Remaining Requests Duplicated date column > Then Home Ribbon > Clicked Date & Time Column > Day > Day of the week
- Navigated to Add column Ribbon > Clicked conditional column > then written If condition > Value 0 equals > Weekend , Else If Value 6 equals > Weekend Else Weekday

3.Remove any extra characters, such as special characters, from the employee ID values. Some IDs may contain a '@' character at the end, which can be cleaned and brought to a common format.

Problem

Extra Characters in the employee_id →

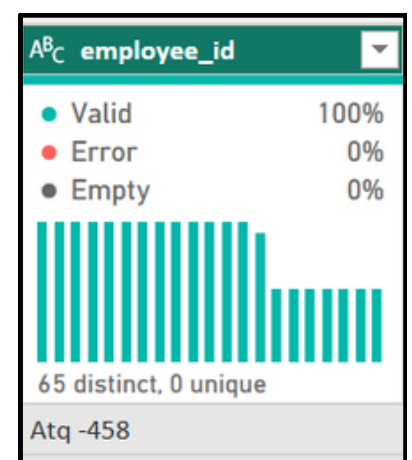


Approach

- First clicked employee_id column > Right clicked > split column > by delimiter > custom > @
- Removed the employee_id.2 because it holds extra characters

Result

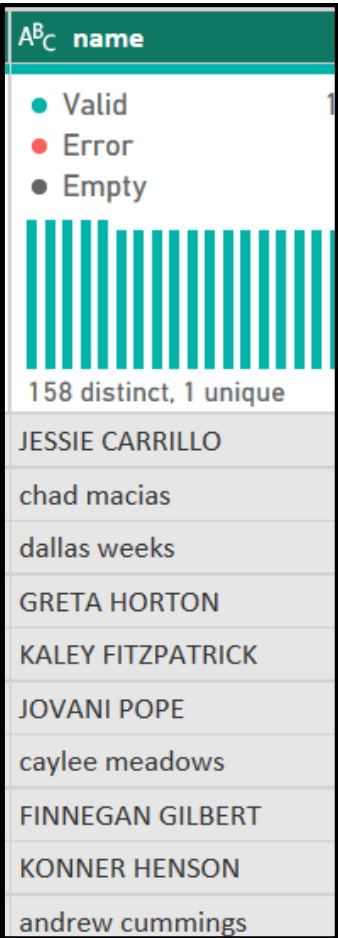
Removed Extra Characters in the employee_id →



4. Standardize the capitalization of names. Convert all names to title case, which means capitalizing the first letter of each word.

Problem

Employee names are not in proper format →

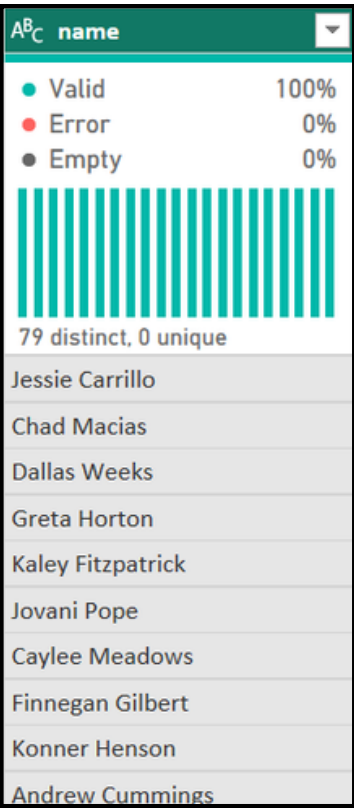


Approach

First clicked name column > Right clicked > Transformed > clicked capitalized by each word

Result

Employee names are in proper format →

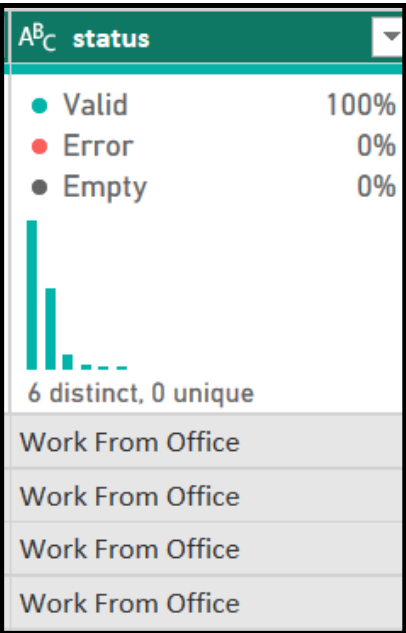


5.Map the corresponding values in the status column with the given abbreviations:

Work From Office --> WFO | Work From Home --> WFH | Birthday Leave --> BL
Menstrual Leave --> ML | Paid Leave --> PL | Sick Leave --> SL | Weekly Off --> WO

Problem

The status is in long text

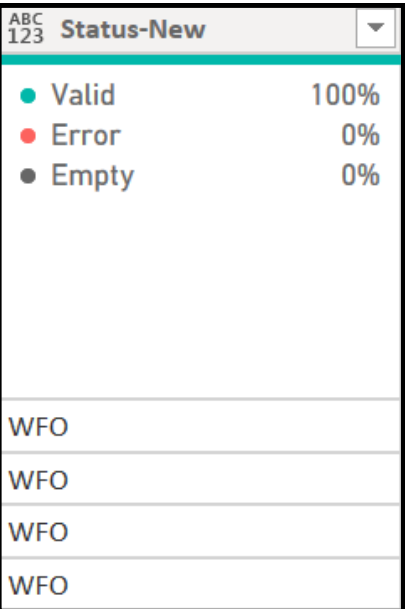


Approach

- Navigated to Add column Ribbon > Clicked conditional column > then written If condition > Value Work From Home equals > WFH,
- Else If Value Work From Office equals > WFO Else Weekday
- Likewise written for all the status

Result

The Status has been changed its abbreviations



02

Data Analysis

4 Real business Questions , Answered with DAX - Once the data was cleaned

Dax measures

1.What is the total count of distinct employee names within the dataset?

Problem

Ensuring clarity on employee identity by analyzing uniqueness in records

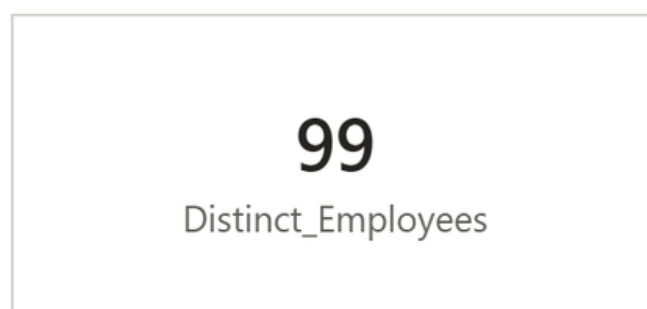
Approach

Dax Measure

```
Distinct_Employees =  
COUNTROWS(  
  
DISTINCT(attendance_data[  
name])  
)
```

Result

Card Visual



2. Calculate the work-from-home (WFH %) percentage in the month of May.

Problem

Assessing remote work adoption to understand workforce flexibility in May.

Approach

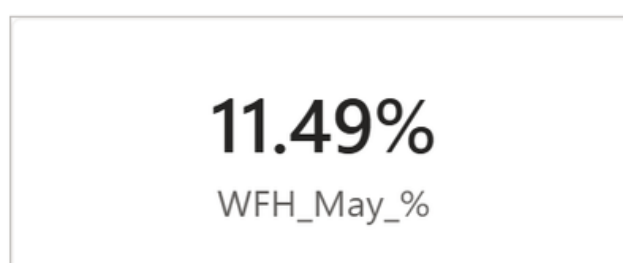
Dax Measure

```
WFH_May_% =  
DIVIDE(  
[WFH_Status_May],  
[Total_present_days_May],0  
)
```

Developed multiple DAX measures to ensure accurate and validated results full implementation available on GitHub Kindly navigate to it.

Result

Card Visual



3.Determine which day of the week had the highest attendance percentage in the month of June.

Problem

Evaluating attendance trends to identify the most engaged weekday in June.

Approach

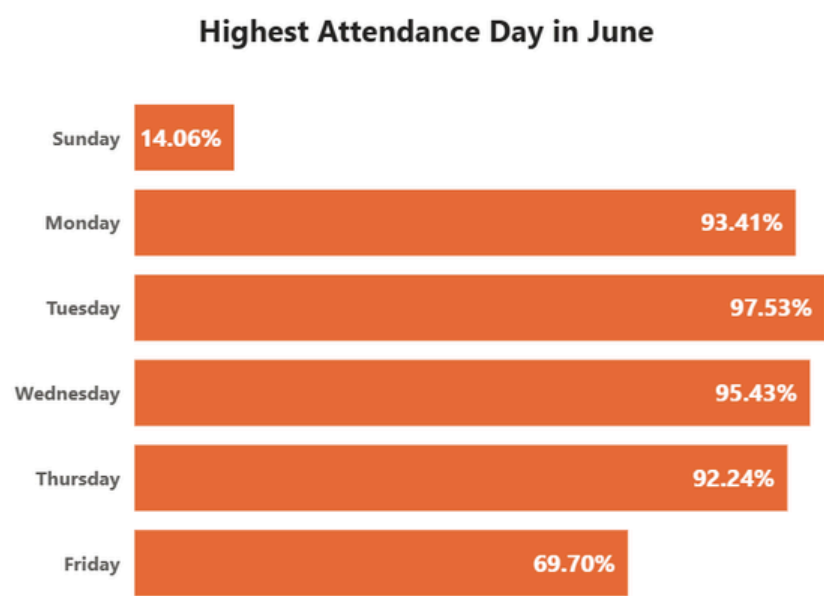
Dax Measure

```
Attendance_%_June =  
DIVIDE(  
[Total_present_days_June],  
[Total_status_June],0  
)
```

Developed multiple DAX measures to ensure accurate and validated results full implementation available on GitHub Kindly navigate to it.

Result (Stacked Bar Chart)

Tuesday has the Highest Attendance Day in June with 97.53 %



4.Find out the number of employees who had a WFH percentage greater than 10% in the month of April.

Problem

Detecting employees with notable remote work patterns in April.

Approach

Dax Measure

```
WFH_April_% =  
DIVIDE(  
[WFH_Status_April],  
[Total_present_days_April],0  
)
```

Developed multiple DAX measures to ensure accurate and validated results full implementation available on GitHub Kindly navigate to it.

Result (Table , Card Visual)

16 employees were having WFH % greater than 10 %

name	WFH_April_%
Andrew Cummings	40.00%
Bo Cordova	21.43%
Caylee Meadows	16.67%
Chad Macias	19.05%
Chris Frye	40.00%
Ella Roth	10.53%
Emma Freeman	15.00%
Gregory Carr	100.00%
Gustavo Ritter	100.00%
Joel Cruz	100.00%
Jonah Hamilton	11.76%
Jovani Pope	31.58%
Kaley Fitzpatrick	30.00%
Maximus Mckenzie	100.00%
Nick Orozco	15.79%
Roman Colon	20.00%

16
WFH_Employee_count

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Github Profile:

<https://github.com/Sachin-Analyst/AtliQ-Virtual-Internship-Week-1>